3211392.IonXpress_018

MTBC functional genotyping report

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Date:	2025-04-22
Name:	3211392.IonXpress_018
Core (v2) ST:	n/a
Species_determination:	n/a
Lineage_number:	1 / 1.1 / 1.1.3
Lineage_name:	Indo-Oceanic
Spoligotype_octal_code:	003070006010060
Resistance_summary:	INH
Resistance_unknown:	AMK / CAP / EMB / FQ / KAN / LFX / LZD / MFL / OFX / RMP / SM
Genotype_unknown:	n/a
MDR_or_XDR:	No
Predicted lineage :	1 Indo-Oceanic 1.1 Indo-Oceanic 1.1.3 Indo-Oceanic
Predicted resistance :	Isoniazid (INH)
Predicted susceptibility :	Bedaquiline (BED) Clofazimine (CLO) Ethionamide (ETH) Para-aminosalicylic acid (PAS) Pyrazinamide (PZA)
Unknown :	Amikacin (AMK) Capreomycin (CAP) Ethambutol (EMB) Fluoroquinolone (FQ) Kanamycin (KAN) Levofloxacin (LFX) Linezolid (LZD) Moxifloxacin (MFL) Ofloxacin (OFX) Rifampicin (RMP) Streptomycin (SM)
Multidrug-resistant :	False
Extensively drug-resistant :	False

Details

Lineage

Mutations lineage

Lineage	Locus	Position	WT	Call	Coverage	Relative coverage (%)	Notes
1 Indo-Oceanic	Rv0524	1104	G	A	17	100.00	
1.1 Indo-Oceanic	Rv3915	1056	G	A	26	100.00	
1.1.3 Indo-Oceanic	Rv1326c	1038	C	T	33	97.06	

Quality metrics I lineage

Locus	QC Passed	Size	Number of bases covered	Average coverage	Start	Stop
Rv0058	Yes	2625	2625	23.64	1	2625
Rv0098	Yes	552	552	26.52	1	552
Rv0284	Yes	3993	3993	18.78	1	3993
Rv0291	Yes	1386	1386	18.80	1	1386
Rv0338c	No	2649	2626	22.24	1	2649
Rv0411c	Yes	987	987	15.07	1	987
Rv0425c	Yes	4620	4620	16.42	1	4620
Rv0450c	Yes	2904	2904	23.41	1	2904
Rv0524	Yes	1389	1389	16.69	1	1389
Rv0668	Yes	3951	3951	25.64	1	3951
Rv0684	Yes	2106	2106	21.68	1	2106
Rv0697	Yes	1440	1440	22.99	1	1440
Rv0781	Yes	711	711	28.90	1	711
Rv0798c	Yes	798	798	22.85	1	798
Rv0835	Yes	645	645	22.77	1	645
Rv0973c	Yes	2004	2004	18.52	1	2004
Rv1013	Yes	1635	1635	23.75	1	1635
Rv1018c	Yes	1488	1488	21.66	1	1488
Rv1111c	Yes	984	984	26.80	1	984
Rv1299	No	1074	975	19.76	32	1074
Rv1324	Yes	915	915	23.96	1	915
Rv1326c	Yes	2196	2196	22.90	1	2196
Rv1332	Yes	657	657	30.38	1	657
Rv1333	Yes	1035	1035	14.39	1	1035
Rv1524	Yes	1245	178	9.19	1	178
Rv1552	Yes	1752	1752	24.75	1	1752
Rv1599	Yes	1317	1317	22.78	1	1317
Rv1617	Yes	1419	1419	20.16	1	1419
Rv1640c	Yes	3519	3519	21.73	1	3519
Rv1661	No	6381	6070	16.23	1	6381
Rv1662	No	4809	4803	15.64	1	4809
Rv2152c	No	1485	1458	16.96	1	1485
Rv2231c	Yes	1095	1095	21.97	1	1095
Rv2343c	Yes	1920	1920	20.25	1	1920
Rv2397c	Yes	1056	1056	28.57	1	1056
Rv2515c	Yes	1248	1248	24.42	1	1248
Rv2555c	Yes	2715	2715	22.41	1	2715
Rv2707	Yes	975	975	21.27	1	975
Rv2907c	Yes	531	531	28.56	1	531
Rv2936	Yes	996	996	21.69	1	996
Rv3029c	Yes	801	801	20.20	1	801
Rv3097c	Yes	1314	1314	19.42	1	1314
Rv3101c	Yes	894	894	31.55	1	894
Rv3111	Yes	513	513	24.73	1	513
Rv3198c	Yes	2103	2103	19.53	1	2103
Rv3336c	Yes	1011	1011	28.89	1	1011
Rv3417c	Yes	1620	1620	24.74	1	1620
Rv3538	Yes	861	861	27.83	1	861
Rv3683	Yes	960	960	21.08	1	960
Rv3708c	Yes	1038	1038	21.48	1	1038
Rv3782	Yes	915	915	22.74	1	915
Rv3794	Yes	3285	3285	21.44	1	3285
Rv3795	Yes	3297	3297	22.17	1	3297
Rv3800c	Yes	5202	5202	21.16	1	5202
Rv3834c	Yes	1260	1260	17.72	1	1260
Rv3843c	Yes	1029	1029	18.24	1	1029
Rv3910	Yes	3555	3555	21.77	1	3555
Rv3915	Yes	1221	1221	26.20	1	1221

Quality metrics II lineage

Locus	QC Passed	Unresolved bases	Start codon	Stop codon	Internal stop codon
Rv0058	Yes	0	Yes	Yes	No
Rv0098	Yes	0	Yes	Yes	No

Locus	QC Passed	Unresolved bases	Start codon	Stop codon	Internal stop codon
Rv0284	Yes	0	Yes	Yes	No
Rv0291	Yes	0	No	Yes	No
Rv0338c	No	0	Yes	Yes	No
Rv0411c	Yes	0	Yes	Yes	No
Rv0425c	Yes	0	Yes	Yes	No
Rv0450c	Yes	0	Yes	Yes	No
Rv0524	Yes	0	Yes	Yes	No
Rv0668	Yes	0	Yes	Yes	No
Rv0684	Yes	0	Yes	Yes	No
Rv0697	Yes	0	Yes	Yes	No
Rv0781	Yes	0	Yes	Yes	No
Rv0798c	Yes	0	Yes	Yes	No
Rv0835	Yes	0	Yes	Yes	No
Rv0973c	Yes	0	Yes	Yes	No
Rv1013	Yes	0	Yes	Yes	No
Rv1018c	Yes	0	Yes	Yes	No
Rv1111c	Yes	0	Yes	Yes	No
Rv1299	No	4	No	Yes	No
Rv1324	Yes	0	Yes	Yes	No
Rv1326c	Yes	0	Yes	Yes	No
Rv1332	Yes	0	Yes	Yes	No
Rv1333	Yes	2	Yes	Yes	No
Rv1524	Yes	0	Yes	No	No
Rv1552	Yes	0	Yes	Yes	No
Rv1599	Yes	0	Yes	Yes	No
Rv1617	Yes	0	Yes	Yes	No
Rv1640c	Yes	0	Yes	Yes	No
Rv1661	No	14	Yes	Yes	No
Rv1662	No	20	Yes	Yes	No
Rv2152c	No	0	Yes	No	No
Rv2231c	Yes	0	Yes	Yes	No
Rv2343c	Yes	0	Yes	Yes	No
Rv2397c	Yes	0	Yes	Yes	No
Rv2515c	Yes	0	Yes	Yes	No
Rv2555c	Yes	0	No	Yes	No
Rv2707	Yes	0	Yes	Yes	No
Rv2907c	Yes	0	Yes	Yes	No
Rv2936	Yes	0	Yes	Yes	No
Rv3029c	Yes	0	Yes	Yes	No
Rv3097c	Yes	0	Yes	Yes	No
Rv3101c	Yes	0	Yes	Yes	No
Rv3111	Yes	0	Yes	Yes	No
Rv3198c	Yes	0	Yes	Yes	No
Rv3336c	Yes	1	Yes	Yes	No
Rv3417c	Yes	0	Yes	Yes	No
Rv3538	Yes	0	Yes	Yes	No
Rv3683	Yes	0	Yes	Yes	No
Rv3708c	Yes	0	Yes	No	No
Rv3782	Yes	0	Yes	Yes	No
Rv3794	Yes	0	Yes	Yes	No
Rv3795	Yes	1	Yes	Yes	No
Rv3800c	Yes	0	Yes	Yes	No
Rv3834c	Yes	0	Yes	Yes	No
Rv3843c	Yes	0	Yes	Yes	No
Rv3910	Yes	0	Yes	Yes	No
Rv3915	Yes	0	Yes	Yes	No

Quality metrics III lineage

Locus	QC Passed	NA mutations	AA mutations	Number of deletions	Number of insertions
Rv0058	Yes	3	1	3	0
Rv0098	Yes	0	0	0	0
Rv0284	Yes	1	1	1	0
Rv0291	Yes	0	0	2	0
Rv0338c	No	3	2	2	0
Rv0411c	Yes	1	0	2	0
Rv0425c	Yes	2	2	0	0
Rv0450c	Yes	1	0	1	0
Rv0524	Yes	1	0	0	0
Rv0668	Yes	4	3	4	0
Rv0684	Yes	0	0	0	0
Rv0697	Yes	1	1	0	0
Rv0781	Yes	0	0	0	1
Rv0798c	Yes	0	0	0	0
Rv0835	Yes	1	0	0	0
Rv0973c	Yes	1	1	1	0

Locus	QC Passed	NA mutations	AA mutations	Number of deletions	Number of insertions
Rv1013	Yes	0	0	0	0
Rv1018c	Yes	1	1	1	0
Rv1111c	Yes	1	1	0	0
Rv1299	No	0	0	2	0
Rv1324	Yes	0	0	0	0
Rv1326c	Yes	4	3	0	0
Rv1332	Yes	0	0	0	0
Rv1333	Yes	0	0	0	0
Rv1524	Yes	0	0	1	0
Rv1552	Yes	2	0	0	0
Rv1599	Yes	0	0	0	0
Rv1617	Yes	0	0	1	0
Rv1640c	Yes	2	1	1	0
Rv1661	No	1	1	11	0
Rv1662	No	2	3	5	0
Rv2152c	No	1	0	3	0
Rv2231c	Yes	0	0	0	0
Rv2343c	Yes	1	0	1	0
Rv2397c	Yes	1	1	0	0
Rv2515c	Yes	0	0	0	0
Rv2555c	Yes	0	0	2	0
Rv2707	Yes	0	0	0	0
Rv2907c	Yes	0	0	0	0
Rv2936	Yes	1	1	2	0
Rv3029c	Yes	0	0	2	0
Rv3097c	Yes	1	1	0	0
Rv3101c	Yes	0	0	0	0
Rv3111	Yes	1	1	0	0
Rv3198c	Yes	0	0	6	0
Rv3336c	Yes	0	0	0	0
Rv3417c	Yes	0	0	0	0
Rv3538	Yes	0	0	0	0
Rv3683	Yes	0	0	1	0
Rv3708c	Yes	2	1	1	0
Rv3782	Yes	0	0	1	0
Rv3794	Yes	2	2	0	0
Rv3795	Yes	1	1	0	0
Rv3800c	Yes	1	0	5	0
Rv3834c	Yes	1	1	1	0
Rv3843c	Yes	2	1	0	0
Rv3910	Yes	3	0	0	0
Rv3915	Yes	1	0	0	0

Resistance

Mutations resistance

Antibiotic	Locus	Codon position	Genomic position	WT	Call	Conclusion	Coverage	Relative coverage (%)	Notes
Ethambutol	embB	378		E	A	Unknown	25	96.15	atypical
Isoniazid	katG	700		S	-	Unknown	5	22.73	atypical

Premature stop codons resistance

Antibiotic	Locus	Codon position	Wild type	Call	Conclusion	Coverage	Relative coverage (%)	Notes
No results								

Insertions/Deletions resistance

Antibiotic	Locus	Codon position	Genomic position	WT	Call	Conclusion	Notes
Isoniazid	katG	37	2156003	G	-	Resistant	

Quality metrics I resistance

Antibiotic	Locus	Gene	QC Passed	Size	Number of bases covered	Average coverage	Start	Stop
	(Rv0565c_promoter)	Rv0565c_promoter	Yes	77	77	18.23	1	77
	(Rv1979c_promoter)	Rv1979c_promoter	Yes	178	178	20.32	1	178
	(Rv2671_promoter)	Rv2671_promoter	Yes	1108	1108	20.36	1	1108
INH	(ahpC_promoter)	ahpC_promoter	Yes	105	105	24.35	1	105
	(ald_promoter)	ald_promoter	Yes	65	65	20.85	1	65
	(alr_promoter)	alr_promoter	Yes	293	293	43.60	1	293
	(atpE_promoter)	atpE_promoter	Yes	48	48	7.21	1	48
	(ddn_promoter)	ddn_promoter	Yes	111	111	21.15	1	111
AMK, KAN	(eis_promoter)	eis_promoter	Yes	139	139	27.42	1	139
EMB	(embA_promoter)	embA_promoter	Yes	85	85	18.52	1	85
	(ethA_promoter)	ethA_promoter	Yes	75	75	30.87	1	75
ETH, INH	(fabG1_promoter)	fabG1_promoter	Yes	140	140	13.66	1	140
	(fbiA_promoter)	fbiA_promoter	Yes	401	401	20.09	1	401
	(fbiC_promoter)	fbiC_promoter	Yes	249	249	19.37	1	249

Antibiotic	Locus	Gene	QC Passed	Size	Number of bases covered	Average coverage	Start	Stop
	(fgd1_promoter)	fgd1_promoter	No	77	68	9.18	1	77
	(gidB_promoter)	gidB_promoter	Yes	131	131	22.42	1	131
	(gyrA_promoter)	gyrA_promoter	Yes	34	34	20.06	1	34
	(gyrB_promoter)	gyrB_promoter	Yes	242	242	9.20	1	242
	(inhA_promoter)	inhA_promoter	Yes	855	855	19.86	1	855
INH	(katG_promoter)	katG_promoter	Yes	37	30	5.00	1	30
	(mmpR_promoter)	mmpR_promoter	Yes	84	84	29.55	1	84
	(mshA_promoter)	mshA_promoter	Yes	47	47	20.49	1	47
	(nat_promoter)	nat_promoter	Yes	251	251	20.15	1	251
	(ndh_promoter)	ndh_promoter	Yes	141	141	31.91	1	141
	(pepQ_promoter)	pepQ_promoter	Yes	33	33	27.33	1	33
PZA	(pncA_promoter)	pncA_promoter	Yes	40	40	28.07	1	40
	(rplC_promoter)	rplC_promoter	Yes	16	16	15.00	1	16
	(rpoB_promoter)	rpoB_promoter	Yes	497	497	22.33	1	497
	(rpsA_promoter)	rpsA_promoter	Yes	162	162	18.20	1	162
	(rpsL_promoter)	rpsL_promoter	Yes	248	248	14.78	1	248
LZD	(rrl)	rrl	No	3138	3108	13.89	1	3138
	(rrl_promoter)	rrl_promoter	Yes	275	275	9.06	1	275
AMK, CAP, KAN, SM	(rrs)	rrs	No	1537	1512	15.55	1	1537
	(rrs_promoter)	rrs_promoter	No	103	82	5.59	1	103
	(thyX_promoter)	thyX_promoter	Yes	243	243	18.62	1	243
CAP	(tlyA_promoter)	tlyA_promoter	Yes	101	101	13.61	1	101
	(ubiA_promoter)	ubiA_promoter	Yes	6	6	17.00	1	6
	(whiB7_promoter)	whiB7_promoter	Yes	429	429	24.78	1	429
FQ, LFX, OFX	Rv0005	gyrB	Yes	2028	2028	22.51	1	2028
FQ, LFX, MFL, OFX	Rv0006	gyrA	Yes	2517	2517	23.45	1	2517
	Rv0341	iniB	Yes	1440	1440	19.31	1	1440
EMB, INH	Rv0342	iniA	Yes	1923	1923	17.75	1	1923
EMB, INH	Rv0343	iniC	Yes	1482	1482	20.35	1	1482
	Rv0407	fgd1	Yes	1011	1011	23.87	1	1011
ETH	Rv0486	mshA	Yes	1443	1443	18.18	1	1443
	Rv0565c	Rv0565c	Yes	1461	1461	15.84	1	1461
RMP	Rv0667	rpoB	Yes	3519	3519	23.93	1	3519
RMP	Rv0668	rpoC	Yes	3951	3951	25.64	1	3951
	Rv0676c	mmpL5	Yes	2895	2895	24.81	1	2895
	Rv0677c	mmpS5	Yes	429	429	24.55	1	429
BED, CLO	Rv0678	mmpR	Yes	498	498	21.42	1	498
SM	Rv0682	rpsL	Yes	375	375	17.33	1	375
LZD	Rv0701	rplC	Yes	654	654	19.61	1	654
	Rv1173	fbiC	Yes	2571	2571	18.97	1	2571
EMB	Rv1267c	embR	Yes	1167	1167	22.27	1	1167
	Rv1305	atpE	Yes	246	246	12.66	1	246
	Rv1483	fabG1	Yes	744	744	20.81	1	744
ETH, INH	Rv1484	inhA	Yes	810	810	28.06	1	810
PZA	Rv1630	rpsA	Yes	1446	1446	22.39	1	1446
CAP	Rv1694	tlyA	Yes	807	807	15.30	1	807
INH	Rv1854c	ndh	Yes	1392	1392	22.65	1	1392
INH	Rv1908c	katG	Yes	2223	2192	17.54	32	2223
	Rv1979c	Rv1979c	Yes	1446	1446	25.27	1	1446
PZA	Rv2043c	pncA	Yes	561	561	30.72	1	561
INH	Rv2245	kasA	Yes	1251	1251	20.80	1	1251
	Rv2416c	eis	Yes	1209	1209	23.14	1	1209
INH	Rv2428	ahpC	Yes	588	588	25.13	1	588
PAS	Rv2447c	folC	Yes	1464	1464	21.53	1	1464
	Rv2535c	pepQ	Yes	1119	1119	20.08	1	1119
PAS	Rv2671	ribD	Yes	777	777	22.29	1	777
	Rv2763c	dfrA	Yes	480	480	16.34	1	480
PAS	Rv2764c	thyA	Yes	792	792	23.79	1	792
	Rv2780	ald	Yes	1116	1116	17.55	1	1116
	Rv3262	fbiB	Yes	1347	1347	19.30	1	1347
	Rv3423c	alr	Yes	1227	1227	22.75	1	1227
	Rv3457c	rpoA	Yes	1044	1044	17.57	1	1044
	Rv3547	ddn	Yes	456	456	16.56	1	456
	Rv3566c	nat	Yes	852	852	20.60	1	852
PZA	Rv3601c	panD	Yes	420	420	17.61	1	420
	Rv3608c	folP1	Yes	843	843	22.72	1	843
	Rv3609c	folE	Yes	609	609	20.53	1	609
	Rv3696c	glpK	Yes	1554	1554	22.94	1	1554
	Rv3792	aftA	Yes	1932	1932	18.92	1	1932
EMB	Rv3793	embC	Yes	3285	3285	24.56	1	3285
EMB	Rv3794	embA	Yes	3285	3285	21.44	1	3285
EMB	Rv3795	embB	Yes	3297	3297	22.17	1	3297
	Rv3806c	ubiA	Yes	909	909	19.63	1	909

Antibiotic	Locus	Gene	QC Passed	Size	Number of bases covered	Average coverage	Start	Stop
ETH	Rv3854c	ethA	Yes	1470	1470	25.48	1	1470
	Rv3855	ethR	Yes	651	651	22.04	1	651
CAP, SM	Rv3919c	gid	Yes	675	675	20.25	1	675

Quality metrics II resistance

Antibiotic	Locus	QC Passed	Unresolved bases	Start Codon	Stop Codon	Internal stop Codon
	(Rv0565c_promoter)	Yes	0			
	(Rv1979c_promoter)	Yes	0			
	(Rv2671_promoter)	Yes	0			
INH	(ahpC_promoter)	Yes	0			
	(ald_promoter)	Yes	0			
	(alr_promoter)	Yes	5			
	(atpE_promoter)	Yes	0			
	(ddn_promoter)	Yes	0			
AMK, KAN	(eis_promoter)	Yes	0			
EMB	(embA_promoter)	Yes	0			
	(ethA_promoter)	Yes	0			
ETH, INH	(fabG1_promoter)	Yes	0			
	(fbiA_promoter)	Yes	0			
	(fbiC_promoter)	Yes	0			
	(fgd1_promoter)	No	0			
	(gidB_promoter)	Yes	0			
	(gyrA_promoter)	Yes	0			
	(gyrB_promoter)	Yes	18			
	(inhA_promoter)	Yes	0			
INH	(katG_promoter)	Yes	0			
	(mmpR_promoter)	Yes	0			
	(mshA_promoter)	Yes	0			
	(nat_promoter)	Yes	0			
	(ndh_promoter)	Yes	0			
	(pepQ_promoter)	Yes	0			
PZA	(pncA_promoter)	Yes	0			
	(rplC_promoter)	Yes	0			
	(rpoB_promoter)	Yes	0			
	(rpsA_promoter)	Yes	0			
	(rpsL_promoter)	Yes	0			
LZD	(rrl)	No	4			
	(rrl_promoter)	Yes	0			
AMK, CAP, KAN, SM	(rrs)	No	0			
	(rrs_promoter)	No	0			
	(thyX_promoter)	Yes	0			
CAP	(tlyA_promoter)	Yes	0			
	(ubiA_promoter)	Yes	0			
	(whiB7_promoter)	Yes	0			
FQ, LFX, OFX	Rv0005	Yes	1	Yes	Yes	No
FQ, LFX, MFL, OFX	Rv0006	Yes	0	Yes	Yes	No
	Rv0341	Yes	0	Yes	Yes	No
EMB, INH	Rv0342	Yes	0	Yes	Yes	No
EMB, INH	Rv0343	Yes	0	Yes	Yes	No
	Rv0407	Yes	0	Yes	Yes	No
ETH	Rv0486	Yes	1	Yes	Yes	No
	Rv0565c	Yes	0	Yes	Yes	No
RMP	Rv0667	Yes	2	Yes	Yes	No
RMP	Rv0668	Yes	0	Yes	Yes	No
	Rv0676c	Yes	1	Yes	Yes	No
	Rv0677c	Yes	0	Yes	Yes	No
BED, CLO	Rv0678	Yes	0	Yes	Yes	No
SM	Rv0682	Yes	0	Yes	Yes	No
LZD	Rv0701	Yes	0	Yes	No	No
	Rv1173	Yes	0	Yes	Yes	No
EMB	Rv1267c	Yes	0	Yes	Yes	No
	Rv1305	Yes	0	Yes	Yes	No
	Rv1483	Yes	0	Yes	Yes	No
ETH, INH	Rv1484	Yes	0	Yes	Yes	No
PZA	Rv1630	Yes	0	Yes	Yes	No
CAP	Rv1694	Yes	0	Yes	Yes	No
INH	Rv1854c	Yes	0	Yes	Yes	No
INH	Rv1908c	Yes	0	Yes	Yes	No
	Rv1979c	Yes	0	Yes	Yes	No
PZA	Rv2043c	Yes	0	Yes	Yes	No
INH	Rv2245	Yes	0	Yes	Yes	No
	Rv2416c	Yes	0	Yes	Yes	No
INH	Rv2428	Yes	0	Yes	Yes	No
PAS	Rv2447c	Yes	0	Yes	Yes	No

Antibiotic	Locus	QC Passed	Unresolved bases	Start Codon	Stop Codon	Internal stop Codon
	Rv2535c	Yes	0	Yes	Yes	No
PAS	Rv2671	Yes	0	Yes	Yes	No
	Rv2763c	Yes	0	Yes	Yes	No
PAS	Rv2764c	Yes	0	Yes	Yes	No
	Rv2780	Yes	0	Yes	Yes	No
	Rv3262	Yes	0	Yes	Yes	No
	Rv3423c	Yes	0	Yes	Yes	No
	Rv3457c	Yes	1	Yes	Yes	No
	Rv3547	Yes	0	Yes	Yes	No
	Rv3566c	Yes	0	Yes	Yes	No
PZA	Rv3601c	Yes	0	Yes	Yes	No
	Rv3608c	Yes	0	Yes	Yes	No
	Rv3609c	Yes	0	Yes	Yes	No
	Rv3696c	Yes	1	Yes	Yes	No
	Rv3792	Yes	5	Yes	Yes	No
EMB	Rv3793	Yes	0	Yes	Yes	No
EMB	Rv3794	Yes	0	Yes	Yes	No
EMB	Rv3795	Yes	1	Yes	Yes	No
	Rv3806c	Yes	0	Yes	Yes	No
ETH	Rv3854c	Yes	0	Yes	Yes	No
	Rv3855	Yes	0	Yes	Yes	No
CAP, SM	Rv3919c	Yes	0	Yes	Yes	No

Quality metrics III resistance

Antibiotic	Locus	QC Passed	NA mutations	AA mutations	Unknown mutations	Number of deletions	Unknown deletions	Number of insertions	Unknown insertions
	(Rv0565c_promoter)	Yes	0		0	0	0	0	0
	(Rv1979c_promoter)	Yes	1		1	0	0	0	0
	(Rv2671_promoter)	Yes	0		0	0	0	0	0
INH	(ahpC_promoter)	Yes	0		0	0	0	0	0
	(ald_promoter)	Yes	1		1	0	0	0	0
	(alr_promoter)	Yes	0		0	0	0	0	0
	(atpE_promoter)	Yes	0		0	0	0	0	0
	(ddn_promoter)	Yes	0		0	0	0	0	0
AMK, KAN	(eis_promoter)	Yes	0		0	0	0	0	0
EMB	(embA_promoter)	Yes	0		0	0	0	0	0
	(ethA_promoter)	Yes	0		0	0	0	0	0
ETH, INH	(fabG1_promoter)	Yes	0		0	0	0	0	0
	(fbiA_promoter)	Yes	0		0	0	0	0	0
	(fbiC_promoter)	Yes	0		0	0	0	0	0
	(fgd1_promoter)	No	0		0	0	0	0	0
	(gidB_promoter)	Yes	0		0	0	0	0	0
	(gyrA_promoter)	Yes	0		0	0	0	0	0
	(gyrB_promoter)	Yes	0		0	0	0	0	0
	(inhA_promoter)	Yes	0		0	0	0	0	0
INH	(katG_promoter)	Yes	0		0	0	0	0	0
	(mmpR_promoter)	Yes	0		0	0	0	0	0
	(mshA_promoter)	Yes	0		0	0	0	0	0
	(nat_promoter)	Yes	0		0	0	0	0	0
	(ndh_promoter)	Yes	0		0	0	0	0	0
	(pepQ_promoter)	Yes	0		0	0	0	0	0
PZA	(pncA_promoter)	Yes	0		0	0	0	0	0
	(rplC_promoter)	Yes	0		0	0	0	0	0
	(rpoB_promoter)	Yes	0		0	0	0	0	0
	(rpsA_promoter)	Yes	0		0	0	0	0	0
	(rpsL_promoter)	Yes	1		1	0	0	0	0
LZD	(rrl)	No	0		0	10	10	0	0
	(rrl_promoter)	Yes	0		0	0	0	0	0
AMK, CAP, KAN, SM	(rrs)	No	0		0	3	3	0	0
	(rrs_promoter)	No	0		0	1	1	0	0
	(thyX_promoter)	Yes	0		0	0	0	0	0
CAP	(tlyA_promoter)	Yes	0		0	0	0	0	0
	(ubiA_promoter)	Yes	0		0	0	0	0	0
	(whiB7_promoter)	Yes	0		0	1	1	0	0
FQ, LFX, OFX	Rv0005	Yes	1	1	1	1	1	0	0
FQ, LFX, MFL, OFX	Rv0006	Yes	6	5	5	1	1	0	0
	Rv0341	Yes	0	0	0	0	0	0	0
EMB, INH	Rv0342	Yes	1	1	1	3	3	0	0
EMB, INH	Rv0343	Yes	0	0	0	0	0	0	0
	Rv0407	Yes	1	0	0	0	0	0	0
ETH	Rv0486	Yes	0	0	0	0	0	0	0
	Rv0565c	Yes	0	0	0	0	0	0	0
RMP	Rv0667	Yes	2	0	0	0	0	0	0
RMP	Rv0668	Yes	4	3	3	4	4	0	0

Antibiotic	Locus	QC Passed	NA mutations	AA mutations	Unknown mutations	Number of deletions	Unknown deletions	Number of insertions	Unknown insertions
	Rv0676c	Yes	2	2	2	0	0	0	0
	Rv0677c	Yes	0	0	0	1	1	0	0
BED, CLO	Rv0678	Yes	0	0	0	0	0	0	0
SM	Rv0682	Yes	0	0	0	0	0	0	0
LZD	Rv0701	Yes	0	0	0	1	1	0	0
	Rv1173	Yes	0	0	0	0	0	0	0
EMB	Rv1267c	Yes	1	1	1	0	0	0	0
	Rv1305	Yes	0	0	0	0	0	0	0
	Rv1483	Yes	0	0	0	0	0	0	0
ETH, INH	Rv1484	Yes	0	0	0	0	0	0	0
PZA	Rv1630	Yes	0	0	0	0	0	0	0
CAP	Rv1694	Yes	1	0	0	0	0	0	0
INH	Rv1854c	Yes	0	0	0	3	3	0	0
INH	Rv1908c	Yes	1	1	1	1	1	0	0
	Rv1979c	Yes	1	1	1	0	0	0	0
PZA	Rv2043c	Yes	0	0	0	0	0	0	0
INH	Rv2245	Yes	1	0	0	0	0	0	0
	Rv2416c	Yes	0	0	0	0	0	0	0
INH	Rv2428	Yes	0	0	0	0	0	0	0
PAS	Rv2447c	Yes	0	0	0	0	0	0	0
	Rv2535c	Yes	0	0	0	1	1	0	0
PAS	Rv2671	Yes	0	0	0	0	0	0	0
	Rv2763c	Yes	0	0	0	0	0	0	0
PAS	Rv2764c	Yes	0	0	0	0	0	0	0
	Rv2780	Yes	0	0	0	1	1	0	0
	Rv3262	Yes	0	0	0	0	0	0	0
	Rv3423c	Yes	1	0	0	2	2	0	0
	Rv3457c	Yes	0	0	0	0	0	0	0
	Rv3547	Yes	0	0	0	0	0	0	0
	Rv3566c	Yes	0	0	0	0	0	0	0
PZA	Rv3601c	Yes	0	0	0	0	0	0	0
	Rv3608c	Yes	0	0	0	0	0	0	0
	Rv3609c	Yes	0	0	0	0	0	0	0
	Rv3696c	Yes	2	1	1	0	0	0	0
	Rv3792	Yes	0	0	0	3	3	0	0
EMB	Rv3793	Yes	3	2	2	0	0	0	0
EMB	Rv3794	Yes	2	2	2	0	0	0	0
EMB	Rv3795	Yes	1	1	1	0	0	0	0
	Rv3806c	Yes	2	1	1	0	0	0	0
ETH	Rv3854c	Yes	0	0	0	0	0	0	0
	Rv3855	Yes	0	0	0	0	0	0	0
CAP, SM	Rv3919c	Yes	3	0	0	0	0	0	0

Genomic variants resistance

Antibiotic	Locus	Gene	Size	Position	Genomic Position	Type	WT	Call	Relative coverage (%)	Strand
	Rv1979c	Rv1979c	1446	857	2222308	Mutation	A (D)	G (G)	82.76	neg
	(Rv1979c_promoter)	Rv1979c_promoter	178	50	2223293	Mutation	A	G	100.00	neg
	(ald_promoter)	ald_promoter	65	34	3086788	Mutation	T	C	100.00	pos
	Rv3423c	alr	1227	168	3841253	Mutation	C (A)	T (A)	100.00	neg
	Rv3423c	alr	1227	1105	3840316	Deletion	G		31.58	neg
	(alr_promoter)	alr_promoter	293	51	3841663	Mutation	G	A	11.11	neg
	(alr_promoter)	alr_promoter	293	52	3841662	Mutation	A	G	53.70	neg
	(alr_promoter)	alr_promoter	293	60	3841654	Mutation	A	C	10.71	neg
	(alr_promoter)	alr_promoter	293	62	3841652	Mutation	A	T	10.71	neg
	(alr_promoter)	alr_promoter	293	89	3841625	Mutation	T	C	41.82	neg
	(alr_promoter)	alr_promoter	293	124	3841590	Deletion	G		27.66	neg
	(alr_promoter)	alr_promoter	293	130	3841584	Mutation	C	A	29.55	neg
	(alr_promoter)	alr_promoter	293	136	3841578	Mutation	G	T	28.26	neg
	(alr_promoter)	alr_promoter	293	140	3841574	Mutation	T	G	26.67	neg
	(alr_promoter)	alr_promoter	293	143	3841571	Deletion	A		23.91	neg
	(alr_promoter)	alr_promoter	293	146	3841568	Mutation	G	A	23.91	neg
EMB	Rv3794	embA	3285	616	4243848	Mutation	G (V)	A (M)	100.00	pos
EMB	Rv3794	embA	3285	956	4244188	Deletion	C		28.57	pos
EMB	Rv3794	embA	3285	2737	4245969	Mutation	C (P)	T (S)	86.36	pos
EMB	Rv3795	embB	3297	133	4246646	Deletion	C		40.00	pos
EMB	Rv3795	embB	3297	1133	4247646	Mutation	A (E)	C (A)	100.00	pos
EMB	Rv3795	embB	3297	2036	4248549	Deletion	C		33.33	pos
EMB	Rv3793	embC	3285	494	4240356	Deletion	C		29.17	pos
EMB	Rv3793	embC	3285	616	4240478	Deletion	G		20.00	pos
EMB	Rv3793	embC	3285	809	4240671	Mutation	C (T)	T (I)	91.18	pos
EMB	Rv3793	embC	3285	1180	4241042	Mutation	A (N)	G (D)	100.00	pos
EMB	Rv3793	embC	3285	1260	4241122	Deletion	C		12.50	pos
EMB	Rv3793	embC	3285	2781	4242643	Mutation	C (R)	T (R)	100.00	pos

Antibiotic	Locus	Gene	Size	Position	Genomic Position	Type	WT	Call	Relative coverage (%)	Strand
EMB	Rv3793	embC	3285	3049	4242911	Deletion	G		31.25	pos
EMB	Rv1267c	embR	1167	329	1417019	Mutation	G (C)	A (Y)	100.00	neg
	Rv1173	fbiC	2571	218	1303148	Deletion	C		33.33	pos
	Rv0407	fgd1	1011	960	491742	Mutation	T (F)	C (F)	93.75	pos
PAS	Rv2447c	folC	1464	772	2746827	Deletion	C		23.81	neg
	Rv3608c	folP1	843	246	4049735	Deletion	C		22.58	neg
CAP, SM	Rv3919c	gid	675	330	4407873	Mutation	G (V)	T (V)	100.00	neg
CAP, SM	Rv3919c	gid	675	423	4407780	Mutation	G (A)	A (A)	100.00	neg
CAP, SM	Rv3919c	gid	675	615	4407588	Mutation	A (A)	G (A)	100.00	neg
	(gidB_promoter)	gidB_promoter	131	87	4408247	Deletion	T		24.00	neg
	Rv3696c	glpK	1554	573	4139183	Mutation	T (G)	G (G)	92.86	neg
	Rv3696c	glpK	1554	803	4138953	Deletion	G		19.35	neg
	Rv3696c	glpK	1554	1025	4138731	Deletion	G		36.67	neg
	Rv3696c	glpK	1554	1379	4138377	Mutation	T (V)	C (A)	100.00	neg
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	61	7362	Mutation	G (E)	C (Q)	100.00	pos
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	284	7585	Mutation	G (S)	C (T)	88.46	pos
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	1151	8452	Mutation	C (A)	T (V)	97.44	pos
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	1842	9143	Mutation	T (I)	C (I)	100.00	pos
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	1975	9276	Mutation	C (L)	G (V)	100.00	pos
FQ, LFX, MFL, OFX	Rv0006	gyrA	2517	2003	9304	Mutation	G (G)	A (D)	100.00	pos
FQ, LFX, OFX	Rv0005	gyrB	2028	873	6112	Mutation	G (M)	C (I)	100.00	pos
EMB, INH	Rv0342	iniA	1923	1443	412280	Mutation	T (H)	G (Q)	87.50	pos
	Rv0341	iniB	1440	50	409411	Deletion	C		33.33	pos
EMB, INH	Rv0343	iniC	1482	320	413076	Deletion	C		20.00	pos
INH	Rv2245	kasA	1251	18	2518132	Mutation	C (T)	T (T)	100.00	pos
INH	Rv2245	kasA	1251	578	2518692	Deletion	C		36.84	pos
INH	Rv1908c	katG	2223	1388	2154724	Mutation	G (R)	T (L)	100.00	neg
INH	Rv1908c	katG	2223	2100	2154012	Deletion	C		22.73	neg
	Rv0676c	mmpL5	2895	737	777744	Deletion	G		27.27	neg
	Rv0676c	mmpL5	2895	2381	776100	Mutation	C (T)	T (I)	100.00	neg
	Rv0676c	mmpL5	2895	2406	776075	Deletion	C		18.92	neg
	Rv0676c	mmpL5	2895	2842	775639	Mutation	A (I)	G (V)	87.50	neg
	Rv0678	mmpR	498	40	779029	Deletion	C		27.27	pos
	Rv0677c	mmpS5	429	256	778650	Deletion	C		20.00	neg
	(nat_promoter)	nat_promoter	251	25	4008409	Deletion	C		23.81	neg
INH	Rv1854c	ndh	1392	217	2102826	Deletion	C		51.72	neg
	Rv2535c	pepQ	1119	956	2859463	Deletion	G		25.00	neg
	Rv3457c	rpoA	1044	196	3878312	Deletion	G		35.00	neg
RMP	Rv0667	rpoB	3519	1677	761483	Deletion	G		26.09	pos
RMP	Rv0667	rpoB	3519	2710	762516	Deletion	G		25.00	pos
RMP	Rv0667	rpoB	3519	3123	762929	Mutation	G (G)	A (G)	95.65	pos
RMP	Rv0667	rpoB	3519	3225	763031	Mutation	T (A)	C (A)	100.00	pos
RMP	Rv0668	rpoC	3951	515	763884	Mutation	C (A)	T (V)	76.19	pos
RMP	Rv0668	rpoC	3951	517	763886	Mutation	C (R)	A (R)	77.27	pos
RMP	Rv0668	rpoC	3951	1780	765149	Deletion	G		27.27	pos
RMP	Rv0668	rpoC	3951	1802	765171	Mutation	C (P)	T (L)	100.00	pos
RMP	Rv0668	rpoC	3951	1861	765230	Mutation	G (A)	A (T)	90.48	pos
RMP	Rv0668	rpoC	3951	2009	765378	Deletion	G		25.00	pos
RMP	Rv0668	rpoC	3951	3149	766518	Deletion	C		26.47	pos
RMP	Rv0668	rpoC	3951	3781	767150	Deletion	G		51.28	pos
	(rpsL_promoter)	rpsL_promoter	248	84	781395	Mutation	T	C	100.00	pos
AMK, CAP, KAN, SM	(rrs)	rrs	1537	494	1472339	Deletion	C		27.78	pos
CAP	Rv1694	tlyA	807	33	1917972	Mutation	A (L)	G (L)	100.00	pos
	Rv3806c	ubiA	909	228	4269606	Mutation	T (R)	C (R)	100.00	neg
	Rv3806c	ubiA	909	447	4269387	Mutation	A (E)	C (D)	100.00	neg

Info

General

Plugin version at time of report: 3.0

Lineage

Determination of the lineage. This feature uses the calculation engine genotyping job results. Results are automatically processed upon retrieving the job.

Analysis Date:	2023-01-05 13:54:00
Plugin version:	3.0
Minimum coverage:	5
Minimum relative coverage (%):	10.0

Resistance

Detection of resistance. This feature uses the calculation engine genotyping job results. Results are automatically processed upon retrieving the job.

Analysis Date:	2023-01-05 13:53:55
Plugin version:	3.0
Minimum coverage:	5
Minimum relative coverage (%):	10
database:	Curated database (version 6.0)

Disclaimer

The contents of this report are for research purposes only and not intended for clinical decision making.